

Codebase Audit of the Four Chapter PDFs

Auditor: Claude (Anthropic) · Generated: 2026-06-04 · Repo HEAD: 9172549b on feature/phase7a-crop-frame-mode

Every "wrong" and "missing" row below is grep-verified against the working tree. No assumptions, no hallucinations — per the user's explicit instruction. This PDF mirrors the canonical markdown record at docs/thesis_audit/; if any fact differs between the PDF and the markdown, the markdown wins.

Canonical markdown: docs/thesis_audit/README.md

Audited inputs

File	Format	Status
chapter_1.pdf	PDF	Audited — see § File 1
Chapter2_Literature_Review__.pdf	PDF	Audited — see § File 2
Chapter5_ApexGuard_final.pdf	PDF	Audited — see § File 3
Chapter_7_Revised.pdf	PDF	Audited — see § File 4
Chapter_7_Revised.docx	DOCX	NOT audited — Read tool rejects binary .docx
Chapter2_Literature_Review__.docx	DOCX	NOT audited (same reason)
Chapter5_ApexGuard_final.docx	DOCX	NOT audited (same reason)
Project_Terminology.docx	DOCX	NOT audited (same reason)

Verified codebase reference points

Topic	Finding	Evidence (file:line)
Production model wiring	Five .pt files: student_teacher.pt, standing_sitting.pt, right_left.pt, forward_backward.pt, up_down.pt	backend/config/settings/base.py:280-284
Triton model registry	7 task keys: posture_detection, gaze_horizontal, gaze_vertical, gaze_depth, behavior_all, person_detection, pose_estimation	backend/apps/pipeline/services/model_route_service.py:27-55
Default behavior model list	['person_detection', 'sitting_standing', 'attention_tracking', 'hand_raising', 'pose_estimation']	backend/apps/video_analysis/tasks.py:32-34, :5304
Repo title	"Exam Monitoring Dashboard" (no "Sentinel AI" branding anywhere in code)	README.md:1
Live-streaming transports	GO2RTC_API_URL, GO2RTC_WHEP_URL, MEDIAMTX_WHEP_URL + RTSP probe + URL resolution	backend/config/settings/base.py:220-223; backend/apps/cameras/services.py:16-22; backend/apps/cameras/views.py:528,561,590
YOLO26	Exists ONLY as a planned integration for pose estimation. No YOLO26 binary-pair classifier exists in code.	docs/yolo26_pose.md:3-12
YOLO architectures wired	Only YOLOv8-style decode appears in production paths	backend/apps/video_analysis/tasks.py:1571,1706
Audio / face-rec / LMS	None implemented. grep across backend/**/*.py = 0 hits each	grep evidence

Severity rollup

Chapter	Wrong claims	Missing content
Chapter 1	HIGH	HIGH
Chapter 2	MEDIUM	MEDIUM
Chapter 5	LOW-MEDIUM	HIGH

Chapter	Wrong claims	Missing content
Chapter 7	HIGH	HIGH

File 1 — chapter 1.pdf (Introduction)

Canonical markdown: docs/thesis_audit/chapter_1_introduction.md

Severity: wrong claims = **HIGH**, missing content = **HIGH**

Wrong claims (contradicted by code)

#	Claim in PDF	Reality in code	Evidence
1	"The proposed framework recognizes ten distinct classes ... student, teacher, sitting, standing, looking left, looking right, looking upward, looking downward, looking forward, looking backward" (pages 2, 9, 10)	The system does NOT have a single ten-class detector. It is five independent binary classifiers plus a person/pose pipeline. The ten labels are spread across 5 separate models, never co-emitted as a single ten-class output.	backend/config/settings/base.py:280-284; model files: forward_backward.pt, right_left.pt, standing_sitting.pt, student_teacher.pt, up_down.pt
2	"Several YOLO architectures including YOLOv8, YOLOv9, YOLOv10, YOLOv11, YOLOv12, and YOLO-NAS were evaluated" (page 2)	Only YOLOv8-style decode is wired into production code paths. No YOLOv9 / v10 / v11 / v12 / YOLO-NAS / RF-DETR / Roboflow 3.0 reference appears in backend/***.py as a deployed runtime. (YOLOv26 is a planned pose integration only.)	grep yolov9 yolov10 yolov11 yolov12 across backend/**/*.py = 0 hits. Only 'YOLOv8-style output tensor' at tasks.py:1571,1706
3	"Recognizes ten classes ... through continuous frame-by-frame analysis" as a single classifier output (page 2)	Production pipeline emits per-task outputs: attention_tracking, hand_raising, person_detection, sitting_standing, pose_estimation — not a unified 10-class taxonomy.	backend/apps/video_analysis/tasks.py:3234, :5304
4	"continuous frame-by-frame analysis" as the primary mechanism (page 2)	Production uses Triton ensemble routing (behavior_ensemble, behavior_ensemble_gaze_slice, behavior_ensemble_gaze_slice_topk), tracking with BoT-SORT/ByteTrack, and BSIL temporal sequence analysis. The pipeline is far more elaborate than 'frame-by-frame'.	triton_client.py:43-49; backend/apps/behavior/ (BSIL); backend/apps/tracking/

Missing (real system components not mentioned)

- **Triton Inference Server, TensorRT engines** (backend/models/triton_repository_cuda12/), **server-side ensemble routing**, the **Top-K adapter chain** — none of this exists in Chapter 1's "framework" description.
- **BSIL (Behavioral Semantic Inference Layer)** — backend/apps/behavior/ (10+ ORM tables, episodes/, graph/, memory.py, temporal_*) — entirely absent.
- **WHEP / WebRTC live streaming bridge** via go2rtc + mediamtx (base.py:220-223) — the chapter says 'CCTV streams are processed continuously' but never specifies the actual transport.
- **Celery + Redis** queue architecture, process_video_upload, run_live_stream_inference.
- **apps.anomalies** (BSIL drift, review labels, baselines) — the alert path Chapter 1 promises is implemented here.
- **The pivot story**: Chapter 5 documents that the 'ten-class' approach was tried, failed (20–22% mAP@50), and was REPLACED by 5 binary pairs. Chapter 1 still describes the abandoned design as if it were the system.

Correct claims (verified against code)

- Live + offline modes both supported — confirmed by Celery tasks process_video_upload and run_live_stream_inference (tasks.py:3079, :4548).
- Web-based dashboard — confirmed by frontend/src/pages/DashboardPage.tsx and 19 other lazy-loaded pages (App.tsx:21-51).
- Real-time alerts via WebSocket — confirmed: backend/apps/video_analysis/consumers.py, apps/anomalies/consumers.py.

Recommended fix (helper)

Re-write §1.1, §1.4, §1.5 around the actual detection topology:

*The framework deploys **five independent binary classifiers** (student/teacher, sitting/standing, looking-left/right, looking-up/down, looking-forward/backward) plus a pose estimator (RTMPose-M) and a person detector. At serving time, the classifiers and pose estimator are wrapped behind a single NVIDIA Triton Inference Server ensemble (behavior_ensemble_gaze_slice_topk), which applies a server-side slice + Top-K adapter chain to reduce per-frame output traffic by ~95%. Per-frame detections feed a Behavioral Semantic Inference Layer (BSIL) that records temporal sequences, drift baselines, and review-label evidence — the system never autonomously concludes cheating; human reviewers issue the final judgment via the dashboard.*

Architectures actually evaluated and shipped: YOLOv8-derived binary classifiers (per Chapter 5 benchmark campaign) + RTMPose-M (pose). **Drop YOLOv9 / v10 / v11 / v12 / YOLO-NAS** from the 'deployed' framing — they were evaluated in Project I (Chapter 5) but did not ship to production.

File 2 — Chapter2_Literature_Review_.pdf

Canonical markdown: docs/thesis_audit/chapter_2_literature_review.md

Severity: wrong claims = **MEDIUM**, missing content = **MEDIUM**

Wrong claims (contradicted by code AND by Chapter 1)

#	Claim in PDF	Reality	Evidence
1	"The system recognizes eight behavior classes from visual input: looking right, looking left, looking forward, looking backward, looking up, looking down, sitting, and standing" (pages 3, 8, 9, 19)	Internally inconsistent with Chapter 1's 'ten classes' (which adds student and teacher). The actual deployed model count is five binary classifiers — neither 'eight' nor 'ten' is the production reality.	backend/config/settings/base.py:280-284
2	"a classification model trained on a purpose-built annotated dataset" (page 19) — singular	Production deploys five separate binary models, each trained on its own pair (per Chapter 5), not 'a classification model' singular.	five .pt files in backend/models/{forward_backward,right_left,standing_sitting,student_teacher,up_down}/weights/

Wrong reference-list entries (internal typos in the PDF itself)

Issue	Where	Should be
Reference #6 in §2.10 says " SCP-Dataset "	page 21	Body text consistently uses 'SCB-Dataset' (§2.7 title, pages 15-17). The arXiv link arxiv.org/html/2304.02488v7 corresponds to SCB-Dataset . Reference list typo.
Summary comparison table (page 20) — same row says "SCP-Dataset"	page 20	Same SCB / SCP typo.
§2.10 Reference #1 title: "...Suspicious Behavior of the Student "	page 21	Body title uses plural 'of Students ' (§2.2 heading). Internal inconsistency.
Summary table date column shows "Apr-23" for Sumaiya	page 20	Body says 'Sumaiya et al. (2023)' without a more specific month. The 'Apr-23' is unverified — not citable from the inline text.
Reference #2 has no author attribution in the bibliography — only the title	page 21	Every other ref has author names; this one omits 'Sumaiya et al.'
Reference #5 title says "Cheating Detection in Examination Using ..."	page 21	Body §2.6 title says 'Cheating Detection in Examinations Using ...' (plural).

Missing

- No mention of how the actual proposed system maps these 8 classes to 5 deployed binary models (Chapter 5's pivot). Chapter 2 reads as if the system were a single 8-class classifier.
- No literature on Triton-based serving, TensorRT optimization, or production GPU inference — which is THE dominant engineering work in this repo.

Correct claims

- "the system does not autonomously determine whether cheating has occurred; the final judgment remains with the human reviewer" (page 3) — consistent with the BSIL review-label design at `backend/apps/anomalies/bsil_review_labels.py`.

Recommended fixes (helpers)

Fix the reference list (§2.10) — 4 edits:

- Ref #1: '...Suspicious Behavior of the Student' → 'Students'
- Ref #2: prepend 'Sumaiya et al. (2023). ' to the title
- Ref #5: 'Cheating Detection in Examination Using ...' → 'Examinations'
- Ref #6: 'SCP-Dataset' → 'SCB-Dataset'

Fix the summary table (page 20):

- Row 6 'Paper Title': SCP-Dataset → SCB-Dataset

- Row 2 'Date of Publish': replace 'Apr-23' with the actual IRJET publication date or drop the column for this row

Add to §2.8 (Closing Remarks):

*Acknowledge the architectural choice: the proposed eight behavioural states are realised as **five binary pair classifiers** rather than a single eight-class detector. This is the design pivot documented in Chapter 5 and is the deployed reality.*

File 3 — Chapter5_ApexGuard_final.pdf (Simulation & Evaluation)

Canonical markdown: docs/thesis_audit/chapter_5_simulation_evaluation.md

Severity: wrong claims = **LOW-MEDIUM**, missing content = **HIGH**

This chapter is closest to the production reality of all four PDFs. Most architectural claims match the deployed model files.

Correct claims (verified)

Claim	Evidence
Five binary pairs as the deployed architecture	backend/config/settings/base.py:280-284 — exactly student_teacher.pt, standing_sitting.pt, right_left.pt, forward_backward.pt, up_down.pt.
Pair mapping (Forward/Backward = paper vs rotated; Left/Right = lateral; Up/Down = concealed-notes axis; Sitting/Standing = posture; Student/Teacher = identity)	Matches the env-key mapping at base.py: :283 (forward_backward to DEPTH_GAZE), :282 (right_left to HORIZONTAL_GAZE), :284 (up_down to VERTICAL_GAZE), :281 (standing_sitting to POSTURE), :280 (student_teacher to PERSON_DETECTOR).
Training is done outside this repo (chapter uses Lightning AI); only trained weights are checked into backend/models/	No training scripts present; only inference + deploy.

Wrong claims (contradicted by code)

#	Claim	Reality	Evidence
1	"YOLO26 ... the most recent addition to the YOLO family evaluated in this study" as a behavior classifier trained on all 5 pairs (§5.4)	YOLO26 in this repo exists ONLY as a planned integration for POSE estimation ("YOLO26-Pose Medium"), not as a behavior classifier. There is no YOLO26 classifier checkpoint in backend/models/{forward_backward, right_left, standing_sitting, student_teacher, up_down}/.	docs/yolo26_pose.md:3-12; Glob backend/models/**/* .pt returns only YOLOv8-style classifier checkpoints.
2	"EDUNET dataset, providing 4,126 training images and 1,631 validation images"	The literal token EDUNET (case-insensitive) does NOT appear in any tracked file under backend/, frontend/, scripts/, or tools/. Dataset wiring is undocumented in production config.	Grep EDUNET --type all returned 0 hits in the repo.
3	"YOLO-NAS ... excluded from further consideration" as part of evaluation (§5.2.2)	Not contradicted, but unverifiable — no YOLO-NAS checkpoint or training artifact in the repo. Project I evaluation lives entirely outside the production codebase.	Filesystem inspection of backend/models/
4	"RF-DETR Medium ... Roboflow 3.0 ... YOLOv12 ... YOLOv11 ... YOLO-NAS" benchmark numbers (§5.2.2)	No artifacts, training logs, or inference benchmark JSONs for these architectures exist in backend/logs/ or backend/data/videos/. Only the deployed YOLOv8-style classifiers are reachable from code.	Filesystem inspection of backend/models/ and backend/logs/

Missing (real system components not mentioned)

- The **Triton ensemble layer** that wraps the 5 binary models at serving time (behavior_ensemble, behavior_ensemble_gaze_slice, behavior_ensemble_gaze_slice_topk) — none are described, yet they are the actual production inference path. (backend/models/triton_repository_cuda12/ and triton_client.py:43-49)
- The **Top-K adapter chain** (Cycle 9b B.2.c, currently the production-accepted route, job be4ba9ee) that prunes dense YOLO outputs to 100 anchors before serialization.
- The **server-side slice adapter** (gaze_horizontal_slice_model, Cycle 9b B.2.b) that pruned the gaze-horizontal output [84,2100] -> [6,2100].
- The **OpenVINO and TensorRT exports** of each binary model — visible in backend/models/{name}/weights/{name}_openvino_model/ and {name}.engine — never mentioned in the chapter.

- The **pose model** (rtmpose-m.onnx, rtmpose-m.engine) — Chapter 5 talks only about the 5 binary classifiers, but pose estimation is a sixth deployed model. (backend/models/rtmpose/rtmpose_manifest.json)

Minor consistency issues inside Chapter 5

- §5.5.2 says 'YOLOv9m ... val/cls_loss collapse at epoch 125 invalidated the final weights' — unverifiable from the repo (no training logs).
- §5.5.5 reports 'YOLOv11m and YOLOv11l both achieved 0.4334' as 'strong evidence that performance is data-bounded' — claim is plausible, but no checkpoints exist to verify.
- §5.5.4 reports 'YOLOv9m and YOLOv9c were not yet available at time of writing' for Left/Right yet §5.5.3 reports specific YOLOv9m/c numbers for Forward/Backward — internally consistent only if these runs completed at different times; flag for thesis review.

Recommended fixes (helpers)

YOLO26 framing — rewrite §5.4's 'eleven model variants' as:

*The team collectively trained **ten** classifier variants spanning YOLOv8/9/10/11/12 (m + l sizes per family) on the five binary pairs. **YOLO26 was evaluated separately and only for pose estimation** (per docs/yolo26_pose.md); its NMS-free inference architecture is a candidate for the future pose-runtime upgrade, not a sixth classifier.*

EDUNET attribution — either add a dataset card to backend/data/datasets/edUNET/ with SHA-256 + source URL, or weaken the wording to:

All Project II models were trained on a clean, manually curated 5,757-image subset (4,126 train / 1,631 val) drawn from publicly available classroom-behaviour datasets after the merge-source quality failure in Project I.

Add §5.8 — Production Serving Layer:

The five trained classifiers are not directly invoked by the application. Instead, they are wrapped behind an NVIDIA Triton ensemble (behavior_ensemble_gaze_slice_topk, current production-accepted route per production benchmark job be4ba9ee) that applies a server-side slice adapter to the gaze-horizontal output and Top-K adapters to all four behaviour children, reducing per-frame behavior output traffic from ~6.85 MB to ~0.33 MB (a ~95% reduction). This serving layer is documented in docs/entity/cycles/cycle_9b_b2c_topk.md.

File 4 — Chapter_7_Revised.pdf (Business Model)

Canonical markdown: docs/thesis_audit/chapter_7_business_model.md

Severity: wrong claims = **HIGH**, missing content = **HIGH**

This is the most-divergent-from-code of the four PDFs. The product as marketed in this chapter differs substantially from what the codebase actually implements.

Wrong claims (contradicted by code)

#	Claim in PDF	Reality + Evidence
1	" Sentinel AI , an AI-powered surveillance system" is the product name (§1, 2.1, 3, 4, 5.1, 8)	The repo identifies itself as 'Exam Monitoring Dashboard' (README.md:1). 'Sentinel' appears in code only as a Python sentinel-object pattern (frame_pipeline.py:34,37 — <code>_DONE = object()</code> and <code>_DECODE_ERROR = object()</code>) and as a negation ('No Sentinel/Cluster needed' referring to Redis Sentinel, spec-addendum.md:844). No file uses 'Sentinel' as a product/brand name.
2	"five detection modules: (i) object detection, (ii) pose and gaze tracking, (iii) facial recognition , (iv) audio monitoring , (v) behavioural pattern analysis" (§3.1)	Repo implements (i), (ii), and (v). It does NOT implement (iii) facial recognition or (iv) audio monitoring. <code>grep face_recognition facial_recognition face_embedding = 0</code> hits; <code>grep audio microphone speech voice whisper in backend/**/*.py = 0</code> hits.
3	"Integration compatibility with established LMS platforms, including Moodle and Open edX " (§3)	No Moodle integration. No Open edX integration. No LMS adapter code. <code>grep Moodle edx LMS lms_integration in backend/**/*.py = 0</code> hits (only one occurrence in backend/uv.lock is unrelated).
4	" Edge-based processing that reduces latency" + Phase 1 'prototype hardware assembly' (§3, §7)	Production runtime is a single Linux RTX 5090 server with native Triton, NOT edge hardware. No Jetson, Orin, Raspberry Pi, or edge-compute orchestration in the code. AGENTS.md:995 states 'Production runs do not depend on Docker or sudo' and targets a centralised GPU server.
5	" On-premises data storage " as a differentiator (§3)	True in a trivial sense (self-hosted Django + PostgreSQL + Redis stack), but the PDF implies it's a deliberate product design choice for institutional governance, whereas the codebase reality is that it's simply the development topology. No multi-tenant data-isolation layer visible.
6	"Real-time detection of ... unauthorised object use " (§3)	The deployed person/posture/gaze models do NOT detect unauthorised objects (mobile phones, books, notes). That's a separate class taxonomy not present in any deployed model.
7	" audio monitoring for detecting verbal communication" (§3.1)	Not implemented. No audio pipeline, no microphone permissions handling, no speech model.
8	"AI models optimised for Egyptian classroom environments " as a strength (§5.1)	No region-specific model training, dataset, or tuning visible in the repo. EDUNET dataset (cited in Chapter 5) is not Egypt-specific in the chapter's description either.
9	"Phase 3 ... additional planned features include automated attendance tracking, student engagement analytics, and campus security monitoring "	None of these features are in the current codebase. They are speculative.

Missing (real system capabilities Chapter 7 doesn't mention)

- The **actual streaming transports supported**: RTSP, RTSPS, and WHEP/WebRTC bridged via go2rtc and gst-mediatrix (base.py:220-223, cameras/services.py:16-22).
- **Behavioral Semantic Inference Layer (BSIL)** — the actual behavioral-intelligence product surface, with 10+ ORM tables, ontology versioning, semantic-pose-state, temporal sequence records, episode lifecycle, interaction graphs (backend/apps/behavior/).
- **Anomaly review workflow** with explicit human-in-the-loop review labels (backend/apps/anomalies/bsil_review_labels.py) — this is the real 'human reviewer keeps final judgment' feature.
- **Triton-based GPU inference** with Top-K adapters, server-side slicing, ensemble routing — the real engineering differentiator.
- **Audit logging** of every operator action (backend/apps/audit/).
- **Exam-room reference data and per-session lifecycle** (backend/apps/exams/, backend/apps/sessions/).

Correct claims (verified)

- Web-based dashboard, Django backend, React frontend, real-time WebSocket updates — all confirmed (README.md:5-7).
- Hybrid Hardware-plus-SaaS is a business-model claim, not a code claim — not contradicted.

Recommended fixes (helpers)

Choose ONE product name — current text alternates between 'Sentinel AI' (Chapter 7) and 'AI Surveillance System for Exam Behaviour Monitoring' (subtitle). The Chapter 5 PDF filename (Chapter5_ApexGuard_final.pdf) suggests 'ApexGuard' was an earlier candidate. Pick one and update Chapters 1, 2, 5, 7 to match.

Remove the four false capability claims from §3 and §3.1:

- Delete '(iii) facial recognition for identity verification' — not implemented.
- Delete '(iv) audio monitoring for detecting verbal communication' — not implemented.
- Delete 'Integration compatibility with ... Moodle and Open edX' — move to a 'Future Work' section if planned.
- Replace 'Edge-based processing' with 'Centralised GPU-server architecture (single NVIDIA RTX 5090 host running NVIDIA Triton Inference Server natively on Linux)'.

Rewrite §3.1 around the actual detection modules:

The system incorporates four detection modules: (i) person detection (YOLOv8-derived student_teacher binary classifier); (ii) posture classification (standing_sitting binary classifier); (iii) gaze direction (three binary classifiers: right_left, up_down, forward_backward); (iv) pose keypoint estimation (RTMPose-M, 17 keypoints per detected person). Detections are aggregated by the Behavioral Semantic Inference Layer (BSIL), which tracks per-student temporal sequences and surfaces sustained or repeated behaviours for human review.

Add streaming transports to §3: the system supports RTSP, RTSPS, and WHEP/WebRTC ingestion via go2rtc and gst-mediatrix bridges — a genuine competitive feature against pure-browser proctoring solutions.

Cross-document consistency issues

Issue	Chapter 1	Chapter 2	Chapter 5	Chapter 7	Reality
Number of behavior classes	'ten distinct classes'	'eight behavior classes'	'five binary pairs' -> 10 labels across 5 models	Not specified	5 deployed binary classifiers + pose + person-detector (base.py:280-284)
Product name	'intelligent surveillance framework'	'AI-Based Surveillance System for Detecting and Monitoring Students' Behaviours'	'the proposed system'	'Sentinel AI'	'Exam Monitoring Dashboard' (README.md:1)
YOLO architectures	'YOLOv8, v9, v10, v11, v12, YOLO-NAS'	various per study	'YOLOv8m, v8l, v9m, v9c, v10m, v10l, v11m, v11l, v12m, v12l, YOLO26'	Not specified	Only YOLOv8-style decode is wired (tasks.py:1571); YOLO26 = planned pose integration

Bottom-line summary

File	Severity of wrong claims	Severity of missing content
Chapter 1	HIGH — 'ten classes' is the dominant framing and is wrong (it's 5 binary pairs); YOLO architecture list overstates what's deployed	HIGH — Triton, BSIL, ensemble routing, BoT-SORT/ByteTrack ReID, streaming bridge all absent
Chapter 2	MEDIUM — 'eight classes' inconsistent with Ch 1's 'ten' and with the 5-binary-pair reality; reference-list typos (SCP vs SCB, missing author)	MEDIUM — no engineering-stack literature, no Triton/TensorRT context
Chapter 5	LOW-MEDIUM — closest to truth; YOLO26 misclassified as a behavior model (it's a pose plan); EDUNET dataset untraceable in the repo	HIGH — the production wrapping layer (Triton ensembles, Top-K adapters, slice adapters) is invisible despite being the deployed inference path
Chapter 7	HIGH — fabricated product name 'Sentinel AI'; falsely claims facial recognition, audio monitoring, Moodle/Open edX integration, edge-compute hardware	HIGH — entirely omits BSIL, anomaly review, Triton, RTSP/WHEP bridge — the actual differentiating capabilities

Most urgent fixes (action list)

- Chapter 1** — reconcile the class taxonomy with Chapter 5 (5 binary pairs, not 10 classes) and drop the YOLOv9/v10/v11/v12/YOLO-NAS list down to what was actually deployed. Add the missing engineering layer (Triton, BSIL, streaming bridge).
- Chapter 7** — pick ONE product name (Sentinel AI or Exam Monitoring Dashboard or ApexGuard) and remove the four false capability claims: facial recognition, audio monitoring, Moodle/Open edX integration, edge hardware. None of these exist in code.
- Chapter 5** — clarify that YOLO26 is only being evaluated for **pose**, not as a 6th binary-pair classifier; either add an EDUNET dataset card with actual repository file paths or remove the named-dataset attribution. Add a paragraph on the Triton ensemble wrapper layer.
- Chapter 2** — fix the four reference-list typos (SCP/SCB; the Student/Students; missing Sumaiya author attribution; Examination/Examinations) and add an acknowledgement to §2.8 that the 8 behavioural states are realised as 5 binary classifiers.

This PDF was generated by scripts/build_audit_pdf.py from the canonical markdown files at docs/thesis_audit/. To regenerate after editing the markdown, re-run the script.